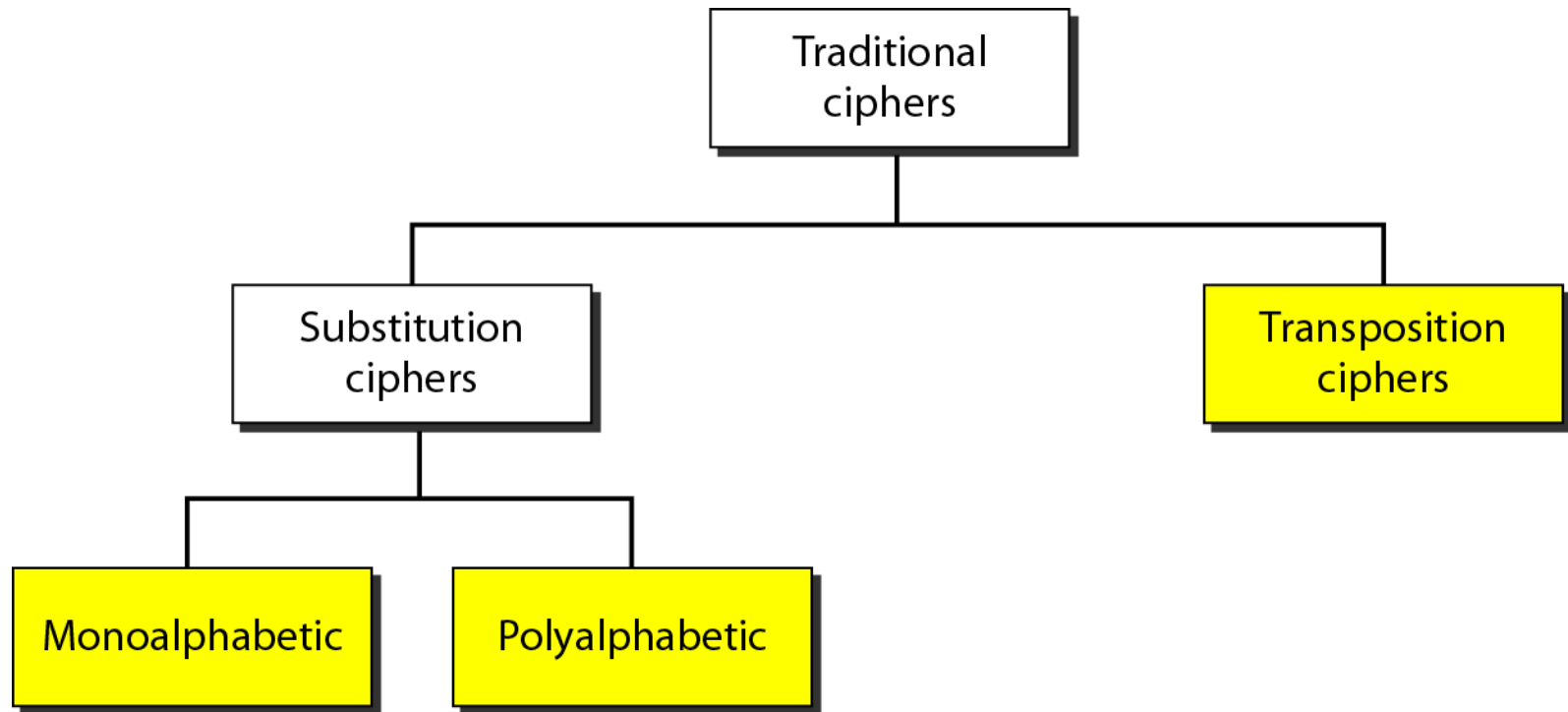


# *Introduction to Cryptography*

# Traditional Ciphers

---



# Substitution Cipher

---

*Note*

**A substitution cipher replaces one symbol with another.**

# Example

---

*The following example gives a plaintext and the equivalent ciphertext. Is it monoalphabetic?*

|                   |
|-------------------|
| Plaintext: HELLO  |
| Ciphertext: KHOOR |

## *Answer*

*This cipher is truly monoalphabetic because both of L's appearances are mapped to O's.*

# Example

---

*The following example gives a plaintext and the equivalent ciphertext. Is it monoalphabetic?*

Plaintext: HELLO

Ciphertext: ABNZF

## *Answer*

*This is not monoalphabetic as each appearance of L is mapped to a different character, N and Z respectively.*

# Shift Cipher

---

*Note*

**The shift cipher is sometimes referred to as the Caesar cipher.**

## Example

---

*Encrypt encrypt the message “HELLO.” using shift cipher with key = 15*

### *Answer*

*We map one character by one character. Each is shifted 15 characters in the alphabet. Letter H is mapped W. Letter E is mapped T. The first L is mapped A. The second L is mapped A too. And O is mapped D. The result is WTAAD.*

# Transposition Cipher

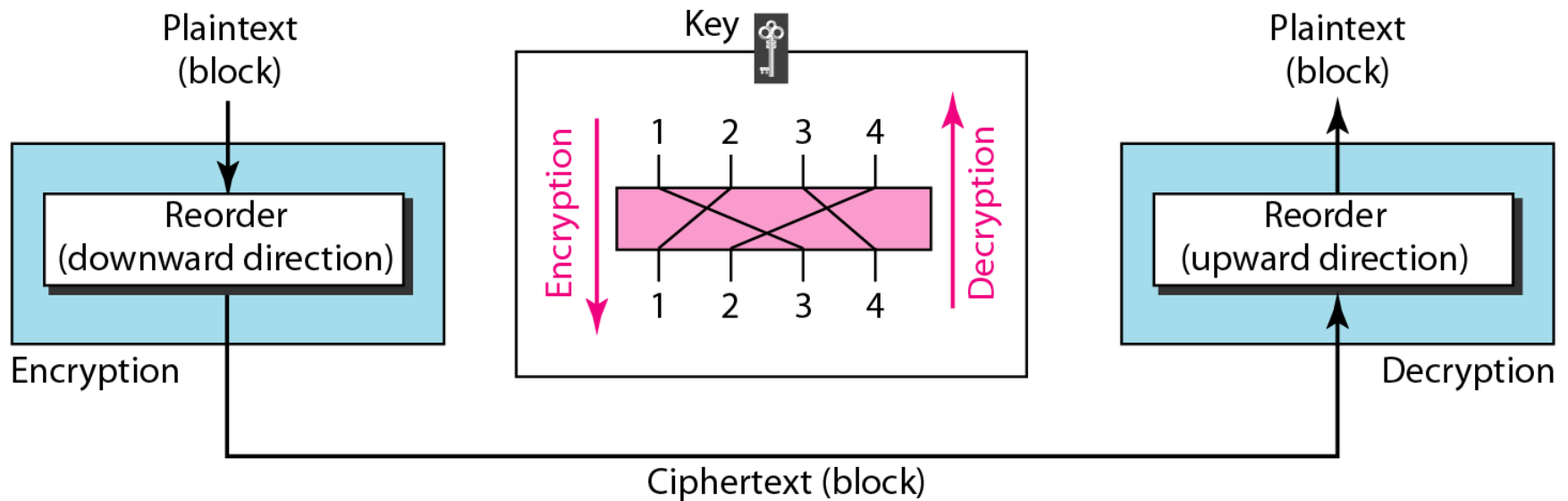
---

*Note*

**A transposition cipher reorders (permutes) symbols in a block of symbols.**



# Transposition Cipher



## Example

---

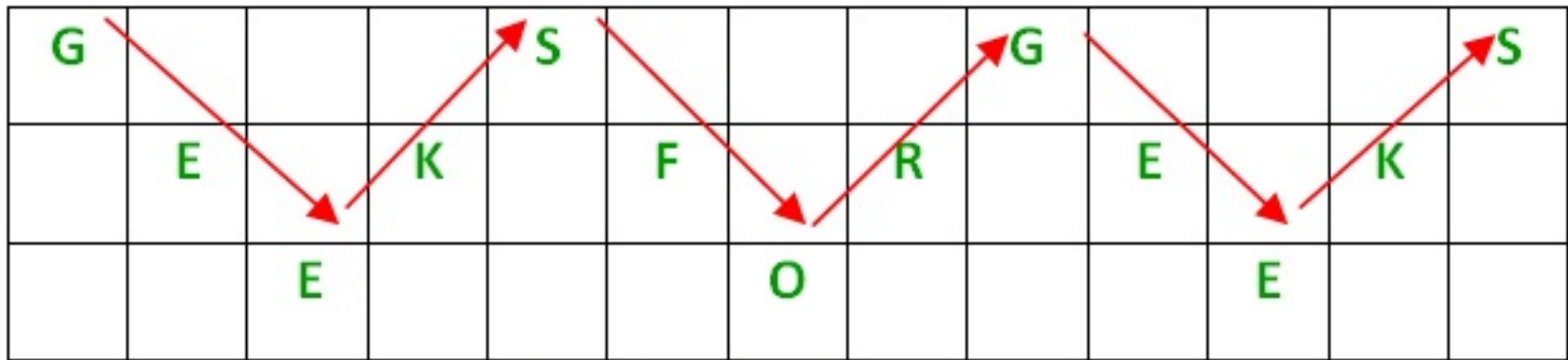
*Encrypt the original message “HELLO MY DEAR,” using the key block given previously.*

### *Answer*

*State by removing the spaces in the original message. Then, divide the message into blocks of four characters. Add a bogus character Z at the end of the third block. The input to the key block is HELL OMYD EARZ. The resultant ciphertext is **ELHLMDOYAZER**.*

# The Rail Fence Encryption Technique

Rail Fence is a kind of a transposition ciphers. It starts by setting the plaintext into a zigzag pattern forming a number of “rails”, as (3) in the following example:



The plaintext is : **GEEKS FOR GEEKS**

Encryption Key : 3 rows

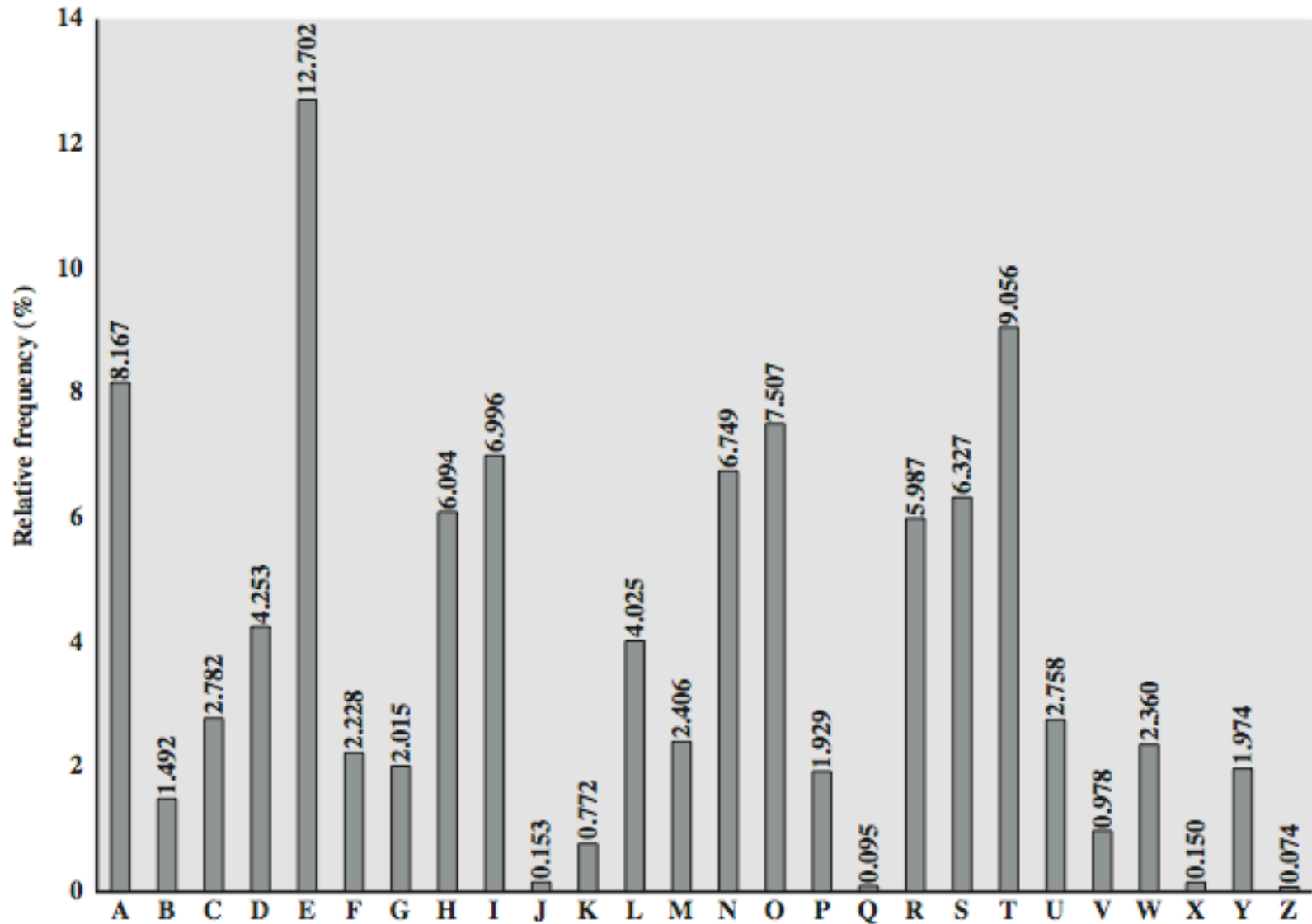
The encrypted text is : **GSGSEKFREKEOE**

# Cryptanalysis

---

- The main target is to recover the key, not just the message,
- General approaches:
  - cryptanalytic attack,
  - brute-force attack
- If any one succeeds in recovering the key, the damage will be catastrophic

# Letter Frequencies (English)



# Example Cryptanalysis

---

➤ **Given the encrypted message:**

UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ  
VUEPHZHMDZSHZOWSFPAPPDTSVPQUZWYMXUZUHSX  
EPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ

➤ **Start by counting relative letter frequencies,**

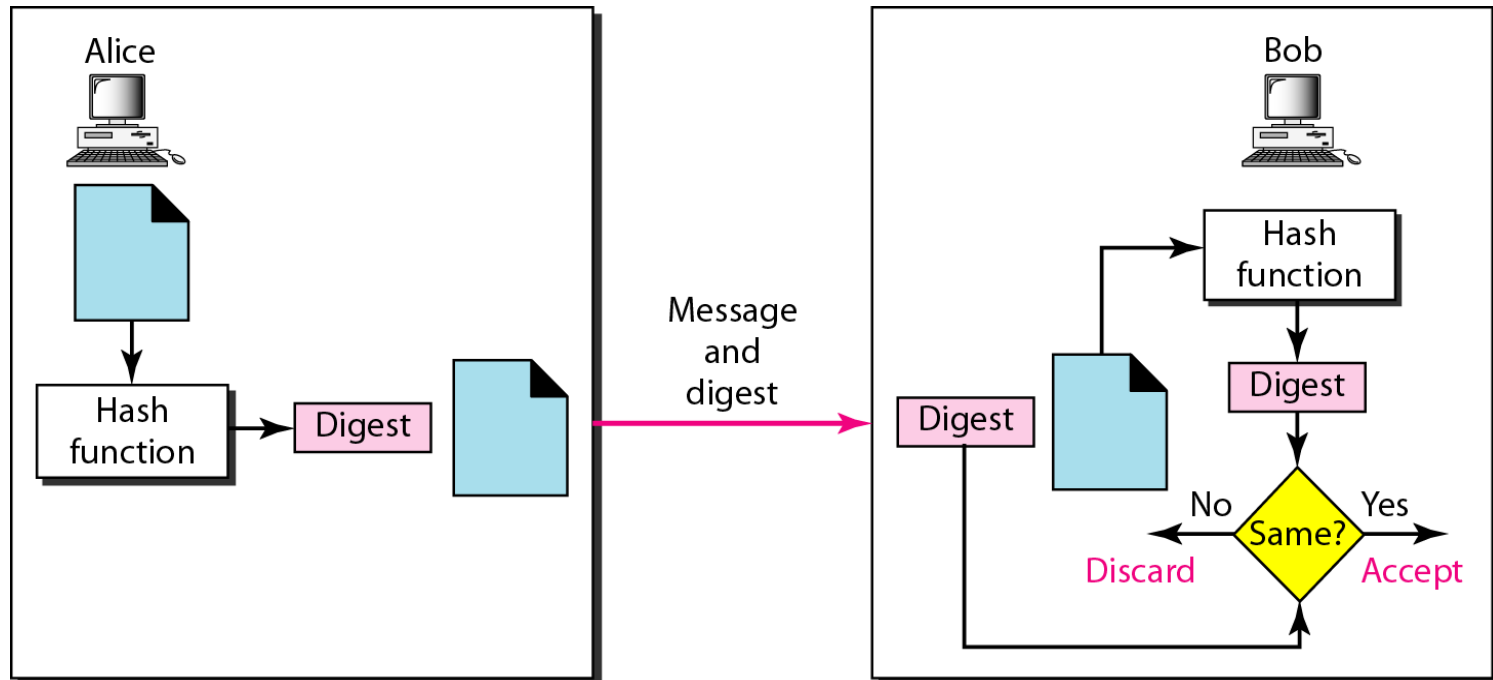
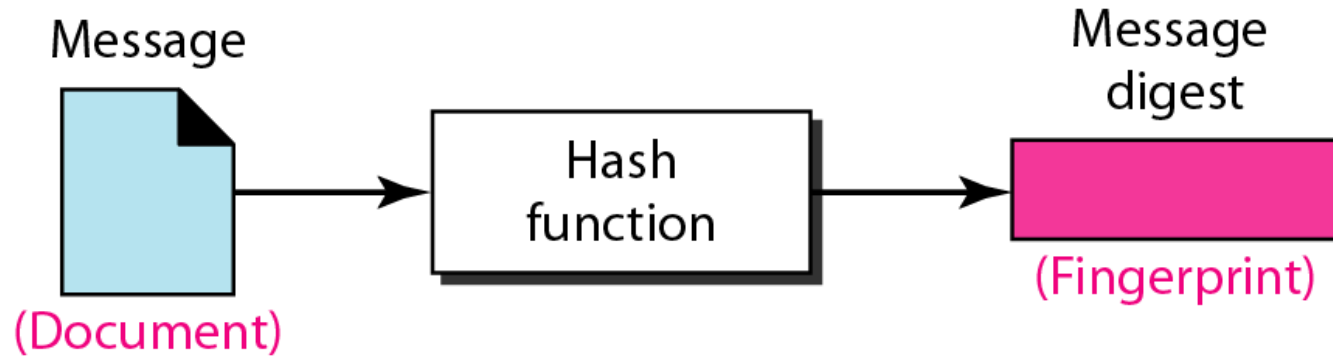
➤ **Then, guess P & Z to be e and t,**

➤ **ZW is th and conclude that ZWP is the,**

➤ **Continue by trial and error, the result is:**

it was disclosed yesterday that several informal but  
direct contacts have been made with political  
representatives of the viet cong in moscow

# Hash Functions



# Criteria of a Hash function ( One-Waynes )

---

*Can we use a conventional lossless compression method as a hashing function?*

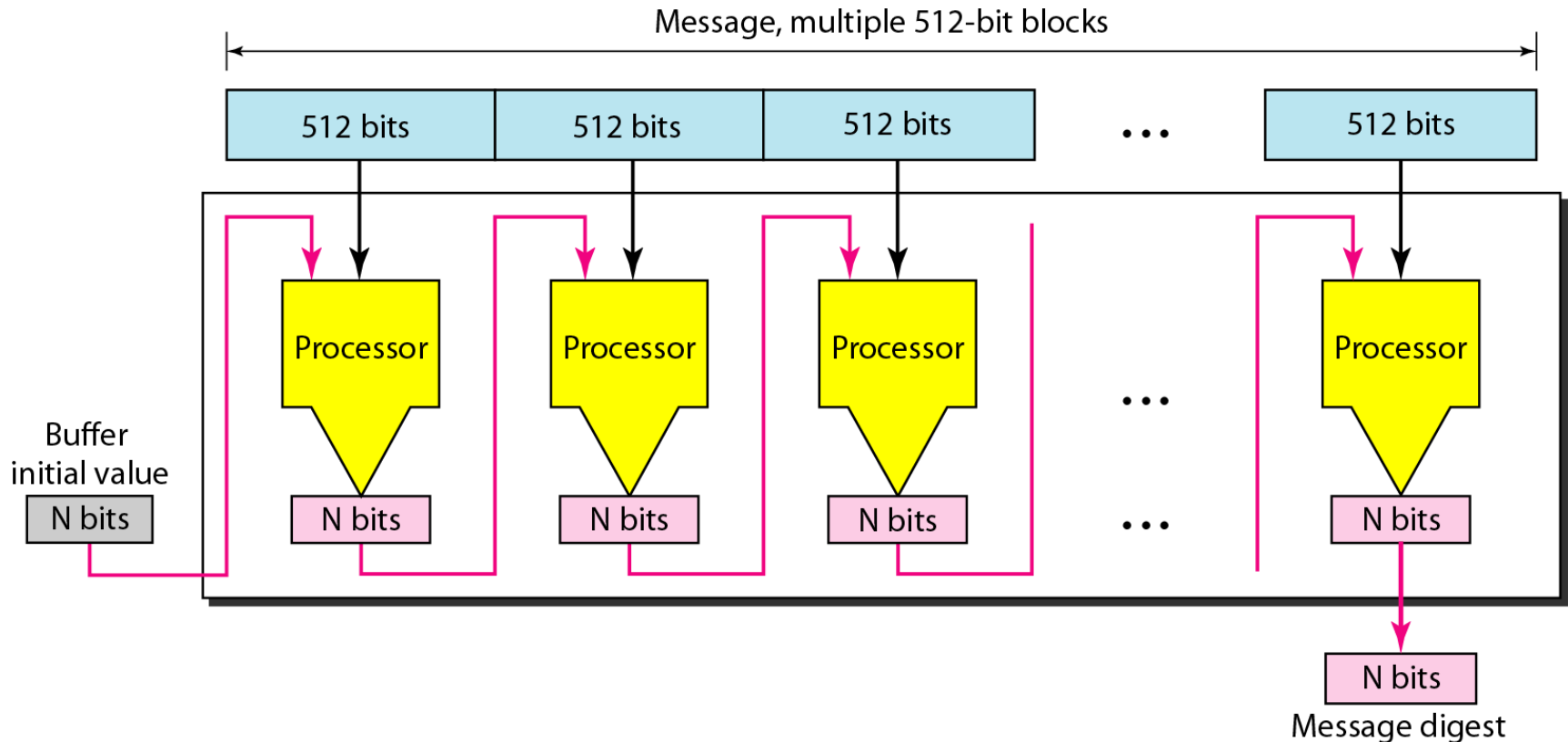
## **Solution**

*We cannot. A lossless compression method creates a compressed message that is reversible. You can uncompress the compressed message to get the original one.*

*A hash function cannot provide authentication. The digest (Hash Value or Hash Code) created by a hash function can detect any modification in the message, but not authentication.*



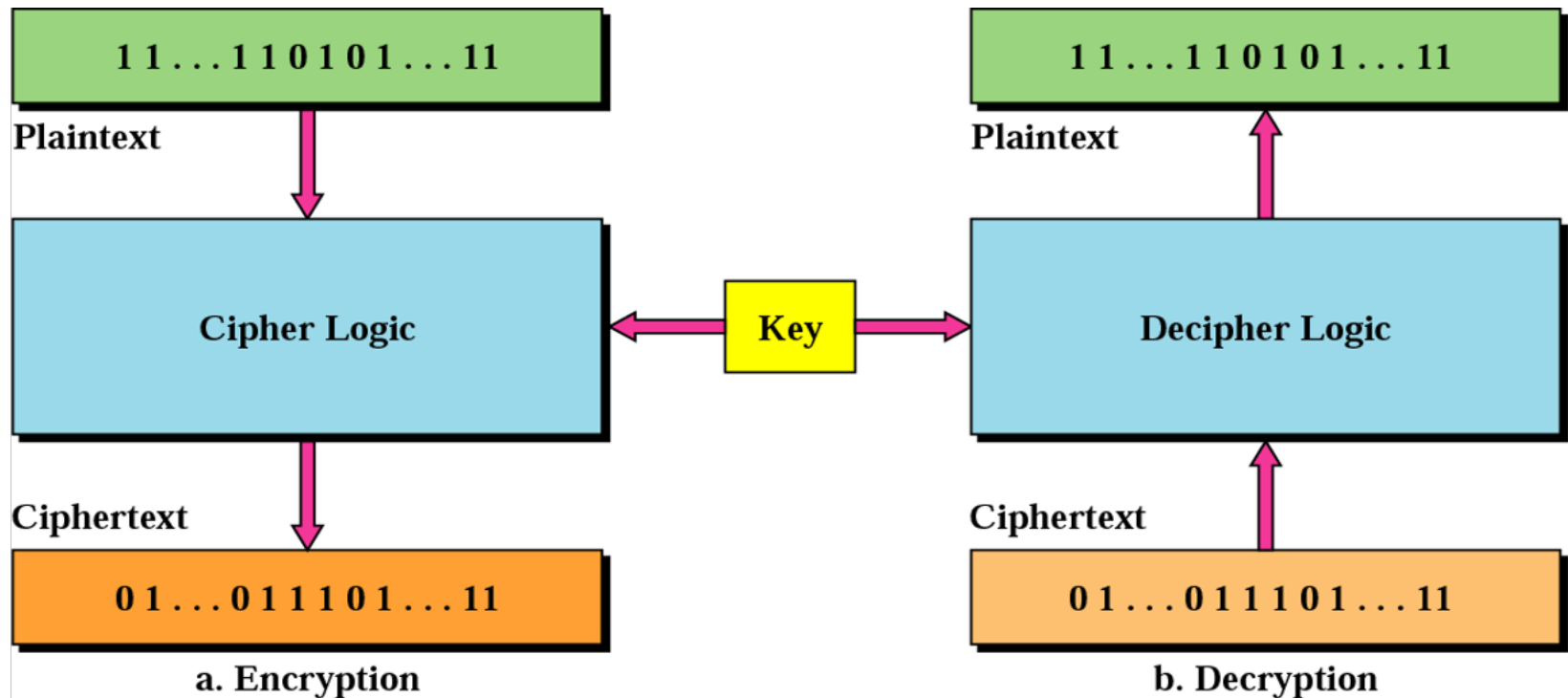
# The Hash Creation Function



**SHA-1 has  
a message digest of 160 bits  
(5 words of 32 bits).**

**MD5 has  
a message digest of 128 bits  
(4 words of 32 bits).**

# Block Cipher : Preventing Crypto Analysis



**The Topic of next week's lecture**

